

Boyu Zhang

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PROFESSIONAL SUMMARY

University of Waterloo Statistics candidate focused on financial data analytics, risk modeling, and digital assets research. Skilled in Python, SQL/MySQL, R, statistical modeling, fixed-income screening, equity research, and market correlation analysis. Experienced turning financial and crypto market datasets into investable insights, risk metrics, and executive-ready reports. Passed CFA Level I.

EXPERIENCE & PROJECTS

Quantitative Strategy Developer (self-employed)

Dec 2024 – Present

- Led a 2-person team to design and productionize a crypto perps strategy across 267 symbols; full Python stack (pandas/NumPy/ccxt/matplotlib) with automated reporting.
- Achieved a 71.8% return, 15.5% max drawdown, and 71% win rate (4.24 Shape Ratio)** across 78 closed trades in last year's in-sample (fee = 5 bps, slip = 0) backtest.
- Conducted walk-forward OOS tests (6- and 3-month, rolling & anchored) showing 66% coverage, 69%/68% win rates over 42 trades, and 40% equity growth consistent across windows.
- Ran cross-section regressions to test which factors predict next-period returns and tracked IC/t-stats over walk-forward windows. Utilized correlation heatmaps/VIF to address collinearity and employed regularized weighting to optimize the strategy.
- <https://vin3ll.github.io/quant-strategy/>

Research Analyst Assistant

Mar 2024 – Aug 2024

- Produced equity research on Pinduoduo (Nasdaq: PDD), Celestica (Nasdaq: CLS), and Tesla (Nasdaq: TSLA) using Wind Financial Terminal, assessing industry structure, fundamentals, catalysts, and business model.
- Built a SQL dataset to organize and screen 10,000+ one-year-to-maturity bonds; analyzed yield patterns and produced shortlists to support fixed-income selection.
- Monitored market dislocations and company-specific catalysts for a real-money portfolio, summarizing entry rationale, downside risks, and post-trade performance.

Research Assistant, Prof. Jawad Addoum, Cornell University

Waterloo, ON

Undergraduate Researcher

May 2025 – July 2025

- Studied corporate finance and applied it to retail leaders Walmart (WMT), Costco (COST), and Amazon (AMZN); computed WACC and asset betas using CAPM and capital-structure measures.
- Analyzed each firm's risk composition and business model based on cost of capital and asset beta, and led the team to deliver a presentation and written report; **ranked 1/20 among the students.**

Research & Publication

Waterloo, ON

Undergraduate Researcher

June 2025 – Aug 2025

- Boyu, Z. (First author). *From Diversifier to Amplifier? Investigating the BTC–NDX Linkage and the Modulating Role of VIX*. Vol. 1 No. 5 (2025); DOI: <https://doi.org/10.61173/03vr3p73> (Published)
- Modeled BTC-NDX dynamic correlation with DCC-GARCH(1, 1) in RStudio**, and log-price visuals from 2017 to 2024.
- Found VIX positively moderates the correlation ($\beta = 0.0106$, $p < 0.001$; $R^2 = 0.301$), connecting crypto-market behavior with volatility regimes.

EDUCATION

Bachelor of Science in Statistics | University of Waterloo | *Waterloo*,

Expected Aug 2026

- Relevant Coursework: Mathematical Statistics, Forecasting, Investment Science and Corporate Finance, Linear Algebra, Linear Models, Experimental Design, Stochastic Process, Derivatives
- President's Scholarship

TECHNICAL SKILLS

Programming & Data: Python, SQL, RStudio, Excel, Wind Financial Terminal

Analytics: regression, time-series analysis, factor analysis, hypothesis testing, backtesting, walk-forward validation, DCC-GARCH

Tools & Languages: Microsoft Office, Mandarin (Native), English (Fluent)

Certificate: Passed CFA Level I